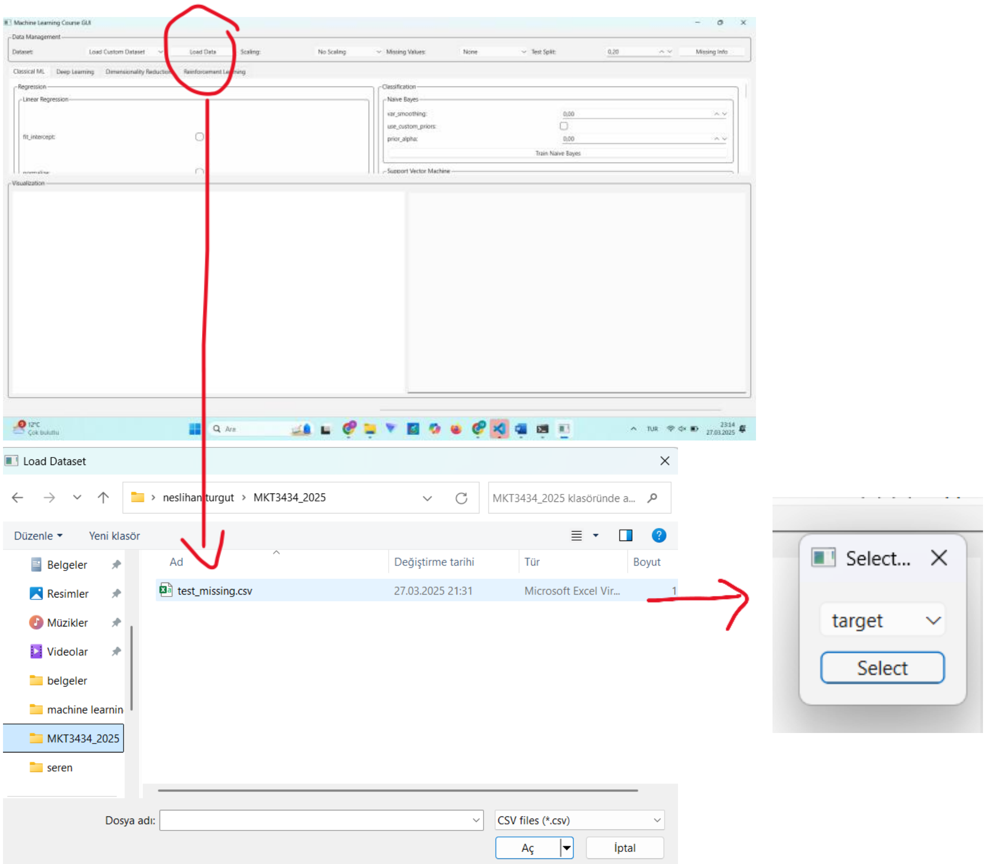
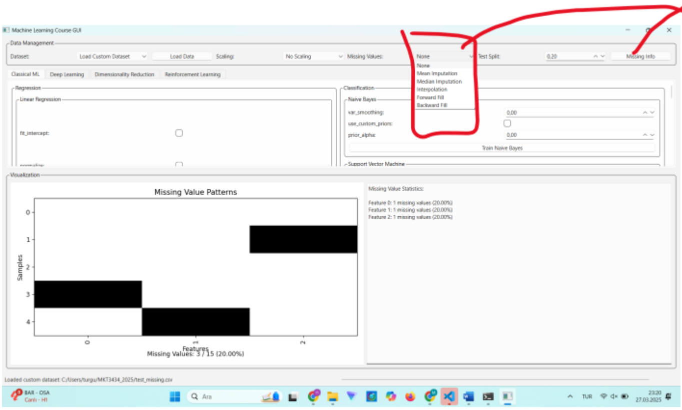
Overall

In this project, we enhanced a machine learning GUI application for the MKT3434 course, focusing on making it more comprehensive and functional. The GUI was implemented using PyQt6, providing a user-friendly interface for loading datasets, preprocessing data with various scaling and missing data handling methods, and training various machine learning models. We added support for a wide range of algorithms including linear regression, SVM, naive Bayes classification, and neural networks, with customizable parameters and configurable loss functions (MSE, MAE, cross-entropy, and hinge loss). The interface offers visualization capabilities for model performance evaluation through interactive plots and comprehensive metrics displays. Additionally, we implemented a comparative missing data handling feature that allows users to benchmark different imputation techniques. The application serves as a practical educational tool for students to experiment with machine learning concepts without writing extensive code, bridging the gap between theoretical understanding and practical implementation.

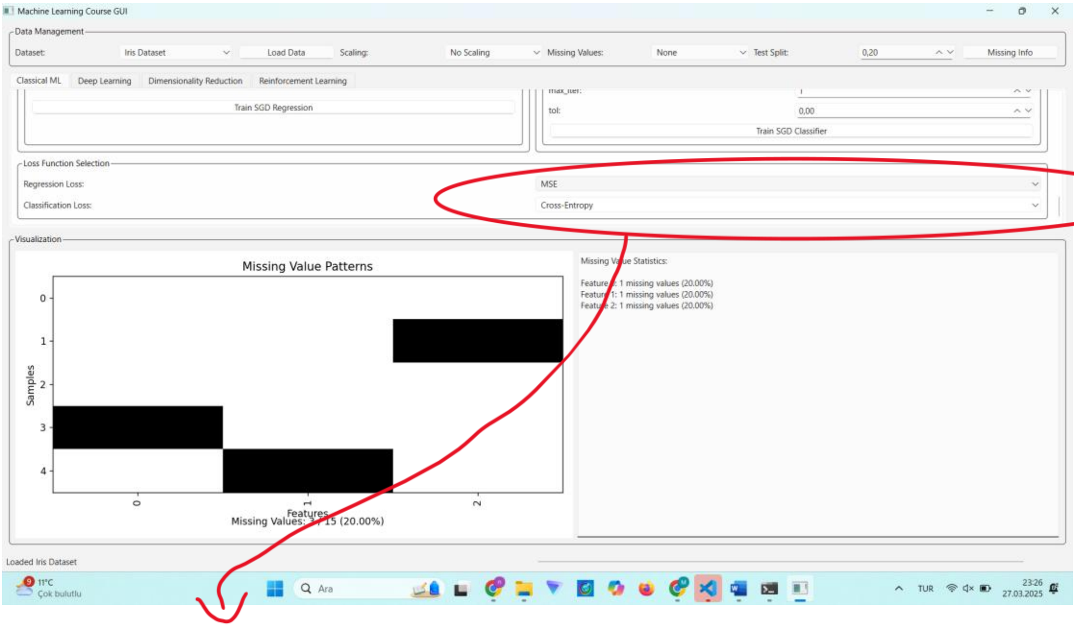
Activated GUİ





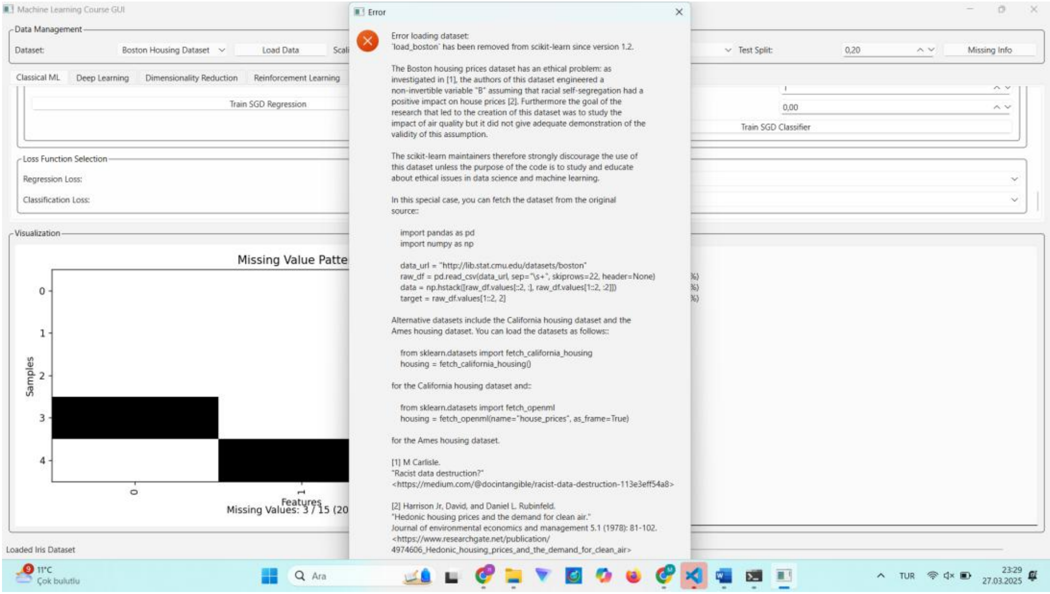
this "missing value" section is not completely correct. due to time constraints, the assignment was completed with a holistic perspective by focusing on the other sections

Iris Dataset



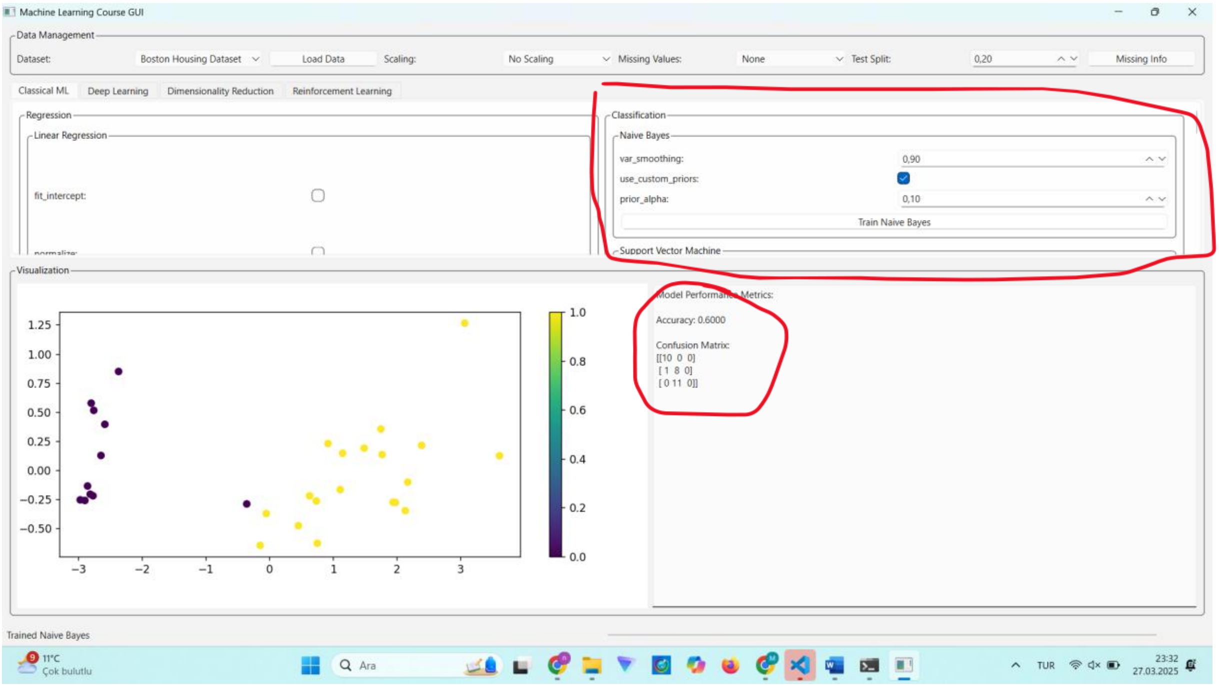
Loss functions for regression and clasificaion have been added.

Boston Housing Dataset



SGD resression is not working properly due to this dataset error.

Classification



Naïve Bayes works properly.